

Linear Regression

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- 1 Forming a Regression Model
 - Understanding the Errors
- 2 Estimators
 - Estimates and Predictions
- 3 Dummy Variables
- 4 Inference in Regression
- 5 Regression “Problems”
 - Include Irrelevant Regressors
 - Heteroscedasticity
 - Multicollinearity
 - Non-linearities
 - Problems Summary

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Linear Factor Models in Finance

Step 1: Time-series regression (estimate factor exposures)

$$r_{i,t} - r_t^f = \alpha_i + \beta_{i,MKT} MKT_t + \beta_{i,SMB} SMB_t + \beta_{i,HML} HML_t + \varepsilon_{i,t}$$

Interpretation: estimate how sensitive asset i is to each systematic risk factor.

Step 2: Cross-sectional regression (estimate prices of risk)

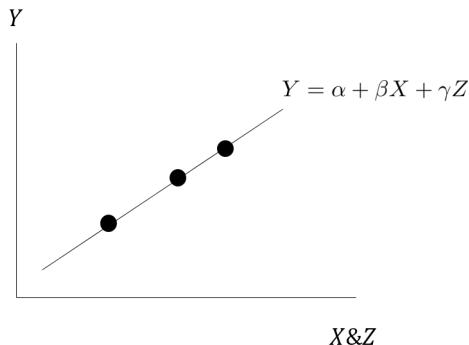
$$r_{i,t} - r_t^f = \lambda_{0,t} + \lambda_{MKT,t} \hat{\beta}_{i,MKT} + \lambda_{SMB,t} \hat{\beta}_{i,SMB} + \lambda_{HML,t} \hat{\beta}_{i,HML} + u_{i,t}$$

Interpretation: determine how much return investors demand for exposure to each factor.

For ease of exposition, in the following slides we will write our toy regression model as

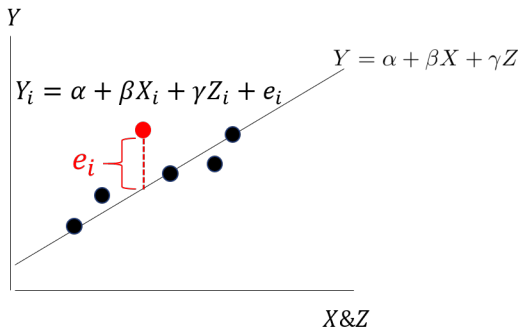
$$Y = \alpha + \beta X + \gamma Z + e$$

A Tautology



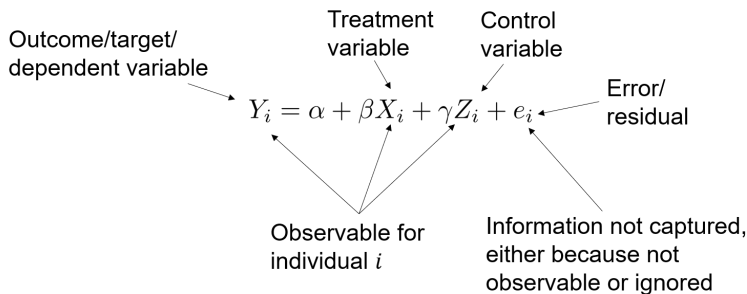
As written above, the model leaves no room for error. Given X and Z we know Y exactly. Every observation follows that equation perfectly, which likely is not realistic.

A model with errors



More than likely there will be some imperfections in the model.
This error or noise is captured by (e).

Population Regression Model



X and Z are regressors / independent variables

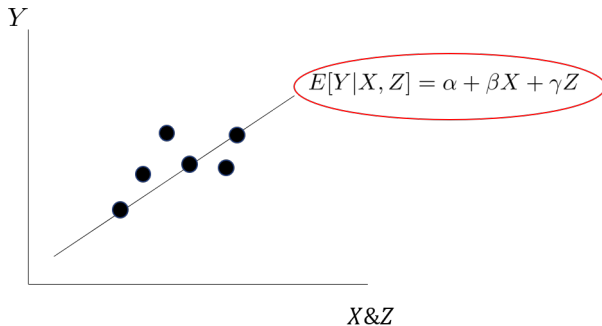
Population Regression Model

Population Regression Model

$$Y_i = \alpha + \beta X_i + \gamma Z_i + e_i$$

Conditional Expectation Function (CEF)

$$E[Y_i|X_i, Z_i] = \alpha + \beta X_i + \gamma Z_i$$



What we are drawing is actually the CEF

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$$Y_i = \alpha + \beta X_i + \gamma Z_i + e_i$$

Q: Two key assumptions about the error term (e_i) A:

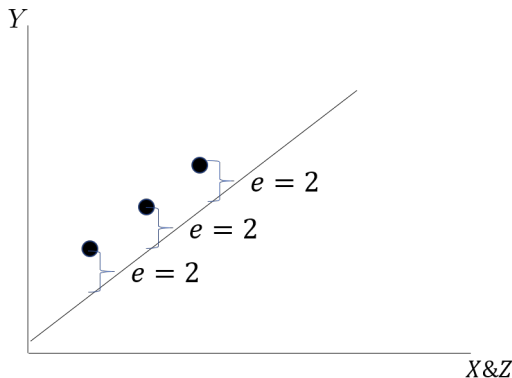
- ① $E[e_i] = 0$
- ② $E[e_i X_i] = E[e_i Z_i] = 0$

Importance of 0 Expected Error

Q: Suppose $E[e_i] = 2$. What is the consequence?

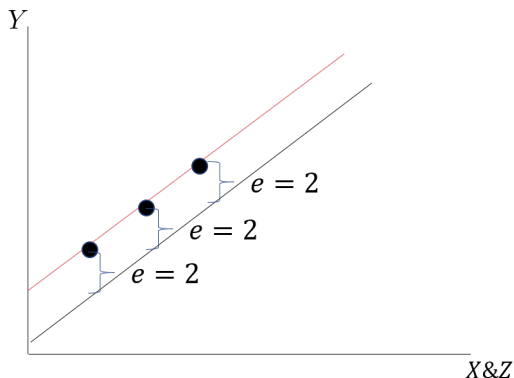
A: $E[Y_i|X_i, Z_i] = \alpha + \beta X_i + \gamma Z_i + 2$

We might have a regression line like this:



Importance of 0 Expected Error

We can reduce the errors by simply shifting the line.



If we have information about our error term, we should include it in our model.

Importance of $E[eX] = 0$

Consider the “orthogonality” assumption $E[e_i X_i] = 0$.

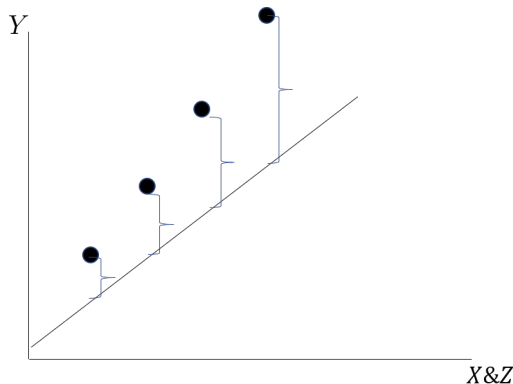
More intuitively, this condition sets the covariance between errors and regressors to 0.

To see this, recall the population Covariance was written as $E[(e - E[e])(X - E[X])]$.

$$\begin{aligned}\text{Cov}(e, X) &= E[(e - E[e])(X - E[X])] \\ &= E[eX - eE[X] - E[e]X + E[e]E[X]] \\ &= E[eX] - E[eE[X]] \\ &= E[eX] - E[e]E[X] \\ &= E[eX]\end{aligned}$$

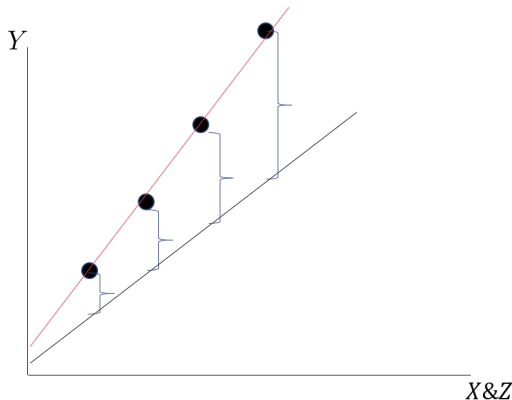
Importance of $E[eX] = 0$

What if the error grew with the regressors (i.e. errors correlated with regressor)?



Importance of $E[eX] = 0$

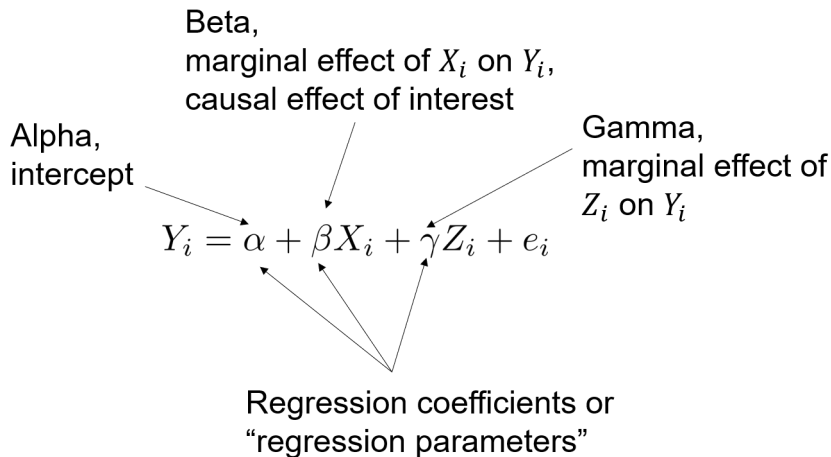
We could reduce the errors by redrawing the regression line.



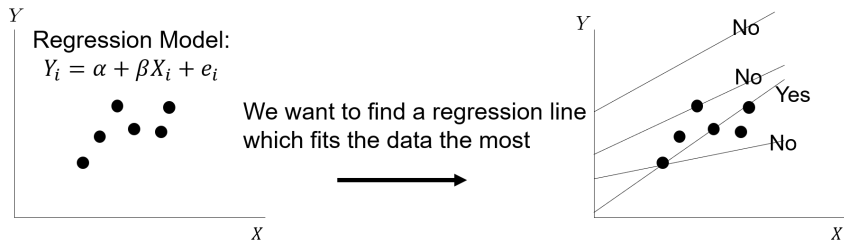
Takeaway: If the errors and regressors were related, then the parameter estimates would have different meanings.

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Estimators



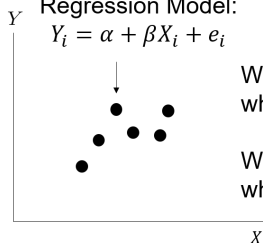
Estimators



Estimators

Regression Model:

$$Y_i = \alpha + \beta X_i + e_i$$



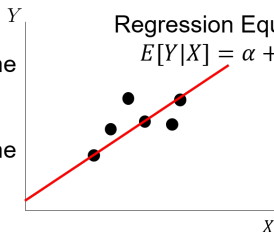
We want to find a regression line which fits the data the most



We want to find a regression line which minimizes the error term

Regression Equation:

$$E[Y|X] = \alpha + \beta X$$



Ordinary Least Squares (OLS)

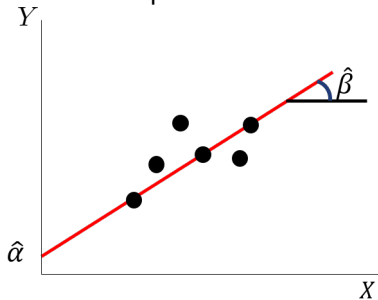
OLS selects the α and β to minimize the sum of squared errors

$$\min \sum_{i=1}^N e_i^2$$

$$\min \sum_{i=1}^N (Y_i - \alpha - \beta X_i)^2$$

$$\hat{\beta} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

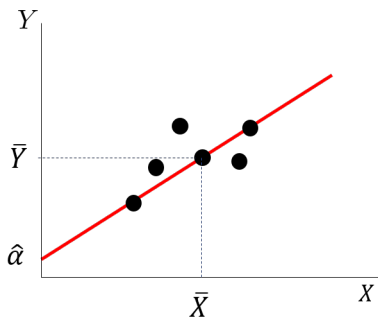


Q: Consequence of squaring the errors?

A: Positive and Negative errors treated equally.

Regression Averages

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X} \rightarrow \bar{Y} = \hat{\alpha} + \hat{\beta}\bar{X}$$



Regression line goes through the averages of dependent / independent variables.

Regression reveals relationships “on average”.

Multivariate Regression

The approach extends easily to multiple regressors.

$$Y_i = \alpha + \beta X_i + \gamma Z_i + e_i$$

$$\min \sum_{i=1}^N e_i^2$$

$$\min \sum_{i=1}^N (Y_i - \alpha - \beta X_i - \gamma Z_i)^2$$

Yields estimated values $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$

Use caution with the “slope” coefficients. Estimators like $\hat{\beta} \neq \frac{\text{Cov}(X,Y)}{\text{Var}(X)}$.

The estimator must be adjusted for the presence of Z . Standard regression packages do this automatically.

Note: There is also a useful geometric interpretation, which views the outcome vector being projected orthogonally on the space spanned by the features.

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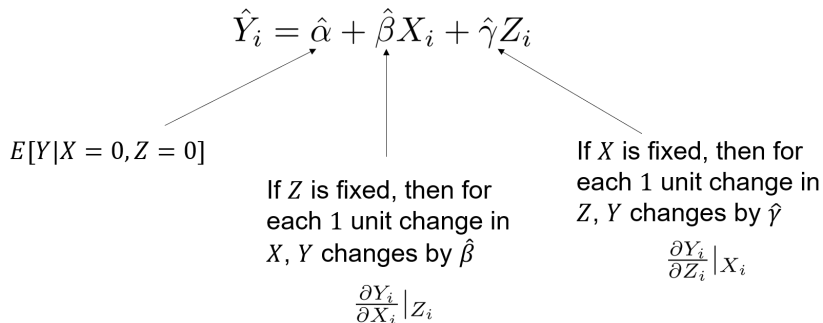
$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$$

Given estimated parameters and knowledge of X and Z , we can estimate the value of Y for any individual i

$$E[Y|X = x, Z = z] = \hat{\alpha} + \hat{\beta}x + \hat{\gamma}z$$

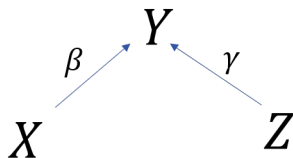
Interpreting Coefficients

Q: How interpret the estimated coefficients in $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$? A:



Interpreting Coefficients

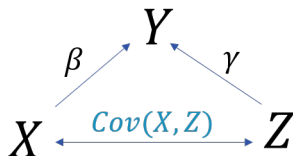
$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$$



The causal relationships between the regressors and Y are indicated by β and γ

Interpreting Coefficients

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$$



If there is a connection between X and Z , β will no longer be a “clean” measure of the connection between X and Y since some of the impact may be channeled through Z .

Caution: Regression is not inherently causal.

Interpreting Coefficients

Suppose the estimated population model is

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$$

But you separate the regressors

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}^A X_i$$

$$\hat{Y}_i = \hat{\alpha} + \hat{\gamma}^B Z_i$$

If $\text{Cov}(X, Z) \neq 0$

$$\hat{\beta}^A \neq \hat{\beta}$$

$$\hat{\gamma}^B \neq \hat{\gamma}$$

Q: Implication?

A: If you forget a regressor/feature/factor, the estimates may be biased.

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Dummies for 1 Way Stratification

$$Y = \alpha + \beta D + e$$

Q: How interpret β ?

A:

- $E[Y|D = 1] = \alpha + \beta$
- $E[Y|D = 0] = \alpha$
- $E[Y|D = 1] - E[Y|D = 0] = \beta$
 - We can interpret β as the difference in the average Y for the two groups ($D=1$ vs $D=0$).

Q: What is the meaning of reject $H_0 : \beta = 0$?

A: Average Y is different across the two groups.

Dummies for 2-way Stratification

$$Y = \alpha + \beta D + \gamma J + e$$

where both D and J are dummy variables.

- $E[Y|D = 1, J = 1] = \alpha + \beta + \gamma$
- $E[Y|D = 1, J = 0] = \alpha + \beta$
- $E[Y|D = 0, J = 1] = \alpha + \gamma$
- $E[Y|D = 0, J = 0] = \alpha$

We can compute β as

- $E[Y|D = 1, J = 1] - E[Y|D = 0, J = 1] = \beta$, or
- $E[Y|D = 1, J = 0] - E[Y|D = 0, J = 0] = \beta$

Q: How interpret β ?

A:

- the (constant) marginal effect of D upon Y (within a given group J)

Dummies for 2-way Stratification

$$Y = \alpha + \beta D + \gamma J + e$$

where both D and J are dummy variables.

- $E[Y|D = 1, J = 1] = \alpha + \beta + \gamma$
- $E[Y|D = 1, J = 0] = \alpha + \beta$
- $E[Y|D = 0, J = 1] = \alpha + \gamma$
- $E[Y|D = 0, J = 0] = \alpha$

We can compute γ as

- $E[Y|D = 1, J = 1] - E[Y|D = 1, J = 0] = \gamma$, or
- $E[Y|D = 0, J = 1] - E[Y|D = 0, J = 0] = \gamma$

Q: How interpret γ ?

- the (constant) marginal effect of J upon Y (within a given group D)

Dummies for Higher Order Stratification

It is possible to extend beyond 2 dummy variables.

$$Y = \alpha + \sum_{k=1}^K \beta_k D_k + e$$

where K is the number of dummy variables included.

K dummy variables implies K -way stratification.

β_k can be interpreted as the marginal effect of D_k upon Y

Interacting 2 Dummy Variables

$$Y = \alpha + \beta D + \gamma J + \theta(D \times J) + e$$

where both D and J are dummy variables.

- $E[Y|D = 1, J = 1] = \alpha + \beta + \gamma + \theta$
- $E[Y|D = 1, J = 0] = \alpha + \beta$
- $E[Y|D = 0, J = 1] = \alpha + \gamma$
- $E[Y|D = 0, J = 0] = \alpha$

We can compute θ as

- $(E[Y|D = 1, J = 1] - E[Y|D = 0, J = 1]) - (E[Y|D = 1, J = 0] - E[Y|D = 0, J = 0]) = (\beta + \theta) - \beta = \theta$

Q: How interpret θ ?

- A: The difference in the marginal effects of D upon Y for groups $J = 1$ and $J = 0$, or...

Interacting 2 Dummy Variables

$$Y = \alpha + \beta D + \gamma J + \theta(D \times J) + e$$

where both D and J are dummy variables.

- $E[Y|D = 1, J = 1] = \alpha + \beta + \gamma + \theta$
- $E[Y|D = 1, J = 0] = \alpha + \beta$
- $E[Y|D = 0, J = 1] = \alpha + \gamma$
- $E[Y|D = 0, J = 0] = \alpha$

We can compute θ as

- $(E[Y|D = 1, J = 1] - E[Y|D = 1, J = 0]) - (E[Y|D = 0, J = 1] - E[Y|D = 0, J = 0]) = (\gamma + \theta) - \gamma = \theta$

In this case we can interpret θ as

- The difference in the marginal effects of J upon Y for groups $D = 1$ and $D = 0$.

Level Effects When Interacting Dummies

$$Y = \alpha + \beta D + \gamma J + \theta(D \times J) + e$$

What if instead we had the following:

$$Y = \alpha + \theta(D \times J) + e$$

- $E[Y|D = 1, J = 1] = \alpha + \theta$
- $E[Y|D = 1, J = 0] = \alpha$
- $E[Y|D = 0, J = 1] = \alpha$
- $E[Y|D = 0, J = 0] = \alpha$

Q: What assumptions are we making by excluding the D and J ?

A: By excluding the “level” effects of D and J we are making the strong claim that these variables don’t impact Y unless these are “switched on” at the same time.

Dummies in the Presence of Continuous Variables

$$Y = \alpha + \beta D + \gamma J + e$$

where D is a dummy and J is continuous.

We can compute β as

- $E[Y|D = 1, J = \bar{J}] = \alpha + \beta + \gamma \bar{J}$
- $E[Y|D = 0, J = \bar{J}] = \alpha + \gamma \bar{J}$
- $E[Y|D = 1, J = \bar{J}] - E[Y|D = 0, J = \bar{J}] = \beta$

Q: How interpret β ?

A:

- The (constant) marginal effect of D upon Y for a given level of J .

Dummies in the Presence of Continuous Variables

$$Y = \alpha + \beta D + \gamma J + e$$

where D is a dummy and J is continuous.

We can compute γ as

- $\frac{\partial Y}{\partial J} = \gamma$

Q: How interpret γ ?

A:

- The (constant) marginal effect of J upon Y .

Interacting a Dummy and Continuous Variables

$$Y = \alpha + \beta D + \gamma J + \theta(D \times J) + e$$

where D is a dummy and J is continuous.

We can explore the role of D as

- $E[Y|D = 1, J = \bar{J}] = \alpha + \beta + \gamma\bar{J} + \theta\bar{J}$
- $E[Y|D = 0, J = \bar{J}] = \alpha + \gamma\bar{J}$
- $E[Y|D = 1, J = \bar{J}] - E[Y|D = 0, J = \bar{J}] = \beta + \theta\bar{J}$

Q: How interpret $\beta + \theta\bar{J}$?

- A: The (constant) marginal effect of D upon Y for a given level of J

Q: How interpret β ?

- A: The (constant) marginal effect of D upon Y when $J = 0$

Interacting a Dummy and Continuous Variables

$$Y = \alpha + \beta D + \gamma J + \theta(D \times J) + e$$

where D is a dummy and J is continuous.

We can compute θ as

- $\frac{\partial Y}{\partial J}|_{D=1} = \gamma + \theta$
- $\frac{\partial Y}{\partial J}|_{D=0} = \gamma$
- $\frac{\partial Y}{\partial J}|_{D=1} - \frac{\partial Y}{\partial J}|_{D=0} = \theta$

Q: How interpret θ ?

- A: The difference in the marginal effect of J upon Y for individuals in group $D = 1$ versus those in group $D = 0$.

Changing Reference Category

Consider again

$$Y = \alpha + \beta D + e$$

where we sort individuals (i.e. observations) into 2 groups: A,B. We then define $D_i = 1$ if i is in Group A, and 0 otherwise (i.e. Group B).

If $\hat{\beta} = 2$ we said expected Y is 2 units higher for Group A than it is for Group B.

Group A is referred to as the “reference” group. Or the “base” group.

We can easily change the reference group by redefining $D_i = 1$ if i is in Group A, and 0 others (i.e. Group A).

A common practice when using dummy variables is to define D such that it takes a value of 1 for the Group of interest.

Dummies for Multiple Categories

Suppose we have 3 groups of interest, rather than 2.

We define

- $D_{1i} = 1$ if i is in Group A, 0 otherwise (i.e. Group B,C)
- $D_{2i} = 1$ if i is in Group B, 0 otherwise (i.e. Group A,C)
- $D_{3i} = 1$ if i is in Group C, 0 otherwise (i.e. Group A,B)

Dummies for Multiple Categories

Our dataset might look like

Obs.	D1	D2	D3
1	0	1	0
2	1	0	0
3	1	0	0
4	0	0	1

Notice that the dummy variables are mutually exclusive.

Notice that when we had 2 groups we only included ONE dummy variable in the regression.

Likewise, when we have 3 groups we only include TWO dummy variables in the regression (we will discuss the reason for this later).

Dummies for Multiple Categories

$$Y = \alpha + \beta_1 D_1 + \beta_2 D_2 + e$$

We can compute α as

- $E[Y|D_1 = 0, D_2 = 0] = \alpha$

Q: How interpret α ?

A:

- Expected Y if we are NOT in Groups A nor B
- Equivalently, α can be interpreted as the expected Y for group C (i.e. the reference group)

Dummies for Multiple Categories

$$Y = \alpha + \beta_1 D_1 + \beta_2 D_2 + e$$

We can compute β_1 as

- $E[Y|D_1 = 1, D_2 = 0] = \alpha + \beta_1$
- $E[Y|D_1 = 0, D_2 = 0] = \alpha$
- $E[Y|D_1 = 1, D_2 = 0] - E[Y|D_1 = 0, D_2 = 0] = \beta_1$

Q: How interpret β_1 ?

A:

- Difference in expected Y for Group A relative to Group C

Dummies for Multiple Categories

$$Y = \alpha + \beta_1 D_1 + \beta_2 D_2 + e$$

We can compute β_2 as

- $E[Y|D_1 = 0, D_2 = 1] = \alpha + \beta_2$
- $E[Y|D_1 = 0, D_2 = 0] = \alpha$
- $E[Y|D_1 = 0, D_2 = 1] - E[Y|D_1 = 0, D_2 = 0] = \beta_2$

Q: How interpret β_2 ?

A:

- Difference in expected Y for Group B relative to Group C

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Testing Significance

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$$

Suppose we are interested in the marginal impact of (X) on (Y).

We want to test $H_0 : \beta = 0$. Suppose $\hat{\beta} = .025$. Is that close enough to 0?

We can apply a t-test to β .

$$H_0 : \beta = 0 \quad H_1 : \beta \neq 0$$

$$t^{stat} = \frac{\hat{\beta} - 0}{se(\hat{\beta})}$$

Reject H_0 if $|t^{stat}| > t^{crit}$

Reject if $p - Value < \alpha$ (Significance Level)

Testing Significance

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i + \hat{\gamma}Z_i$$

We can easily extend the testing procedure to values of β other than 0.
Perhaps we want to test $H_0 : \beta = 1$. Q: How augment t-test of β ?

$$H_0 : \beta = 1 \quad H_1 : \beta \neq 1$$

$$t^{stat} = \frac{\hat{\beta} - 1}{se(\hat{\beta})}$$

Reject H_0 if $|t^{stat}| > t^{crit}$

Reject if $p - Value < \alpha$ (Significance Level)

Standard Error of β

$$se(\hat{\beta}) = \frac{\hat{\sigma}_e}{\sqrt{SST_X(1 - R_X^2)}}$$

where

- $\hat{\sigma}_e = \sqrt{\frac{1}{N-K} \sum_{i=1}^N (Y_i - \hat{\alpha} - \hat{\beta}X_i - \hat{\gamma}Z_i)^2}$
- N is the sample size
- K is the number of parameters to be estimated ($K=3$ in this case)
- $SST_X = \sum_{i=1}^N (X_i - \bar{X})^2$
- R_X^2 is the goodness of fit measure from the auxiliary regression $X_i = \theta_0 + \theta_1 Z_i + u_i$ computed as $R_X^2 = 1 - \frac{\sigma_u^2}{\sum_{i=1}^N (X_i - \bar{X})^2}$

Standard Error of β

$$se(\hat{\beta}) = \frac{\hat{\sigma}_e}{\sqrt{SST_X(1 - R_X^2)}}$$

Rewriting slightly provides an intuitive decomposition

$$se(\hat{\beta}) = \hat{\sigma}_e \times \frac{1}{\sqrt{N}} \times \frac{1}{\sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}} \times \frac{1}{\sqrt{(1 - R_X^2)}}$$

Standard Error of β and Type 1 Error

$$se(\hat{\beta}) = \hat{\sigma}_e \times \frac{1}{\sqrt{N}} \times \frac{1}{\sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}} \times \frac{1}{\sqrt{(1 - R_X^2)}}$$

- Q: How does Type 1 error change with the components of this std error? (trick question)
- A:
 - It doesn't
 - Type 1 Error = $\Pr(\text{Reject Null} \mid \text{Null is True})$
 - This is fixed by the analyst at their desired significance level.

Standard Error of β and Type 2 Error

$$se(\hat{\beta}) = \hat{\sigma}_e \times \frac{1}{\sqrt{N}} \times \frac{1}{\sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}} \times \frac{1}{\sqrt{(1 - R_X^2)}}$$

Q: How does Type 2 error change with the components of this std error?

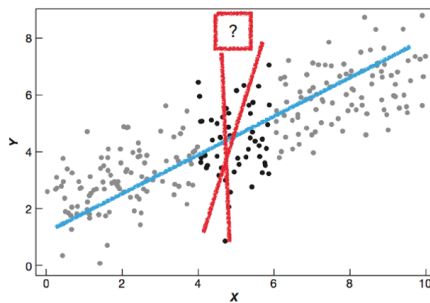
A: Recall Type 2 error = $\Pr(\text{Fail to Reject Null} \mid \text{Null is False})$

- $\hat{\sigma}_e \uparrow \rightarrow se(\hat{\beta}) \uparrow \rightarrow t^{stat} \downarrow \rightarrow \text{Type 2 Error} \uparrow$
- $N \uparrow \rightarrow se(\hat{\beta}) \downarrow \rightarrow t^{stat} \uparrow \rightarrow \text{Type 2 Error} \downarrow$
- $R_X^2 \uparrow \rightarrow se(\hat{\beta}) \uparrow \rightarrow t^{stat} \downarrow \rightarrow \text{Type 2 Error} \uparrow$
- $\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2 \uparrow \rightarrow se(\hat{\beta}) \downarrow \rightarrow t^{stat} \uparrow \rightarrow \text{Type 2 Error} \downarrow$

Let's visualize the impact of dispersion of X

Cross Sectional Dispersion

$$se(\hat{\beta}) = \hat{\sigma}_e \times \frac{1}{\sqrt{N}} \times \frac{1}{\sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2}} \times \frac{1}{\sqrt{(1 - R_X^2)}}$$



SST_X would be small, which means $se(\hat{\beta})$ would be large.

This implies that the t^{stat} would be smaller

Raising the probability of “accepting” the null and hence a higher Type 2 error
 $Pr(\text{Accept } H_0 | H_0 \text{ False})$

Consider the black dots. Q: What if X had small spread of data?

Standard Error of β

This formulation $se(\hat{\beta}) = \frac{\hat{\sigma}_e}{\sqrt{SST_X(1-R_X^2)}}$ is consistent with 2.15 in Mastering Metrics.

$$\begin{aligned} se(\hat{\beta}) &= \frac{\hat{\sigma}_e}{\sqrt{SST_X(1-R_X^2)}} = \frac{\hat{\sigma}_e}{\sqrt{SST_X - SST_X R_X^2}} \\ &= \frac{\hat{\sigma}_e}{\sqrt{SST_X - SST_X(1 - \frac{\sum u^2}{\sum (X-\bar{X})^2})}} \end{aligned}$$

where u is residual from regression of X on all other regressors

$$= \frac{\hat{\sigma}_e}{\sqrt{\sum u^2}} = \frac{\hat{\sigma}_e}{\sqrt{N\hat{\sigma}_u^2}} = \frac{\hat{\sigma}_e}{\sqrt{N}} \times \frac{1}{\hat{\sigma}_u}$$

Testing a Linear Combination of Parameters

Consider

$$Y = \alpha + \beta X + \gamma Z + e$$

Suppose we want to test the equality $\beta = 2\gamma$. Then we have

$$H_0 : \beta = 2\gamma \Rightarrow \beta - 2\gamma = 0$$

$$H_1 : \beta - 2\gamma \neq 0$$

The t-test statistic is constructed and evaluated similar to usual

$$t^{stat} = \frac{\hat{\beta} - 2\hat{\gamma}}{se(\hat{\beta} - 2\hat{\gamma})}$$

Note that the denominator is built from

$Var(\hat{\beta} - 2\hat{\gamma}) = Var(\hat{\beta}) + 4Var(\hat{\gamma}) - 4Cov(\hat{\beta}, \hat{\gamma})$. The covariance term will be produced automatically by any decent econometrics package.

Testing Multiple Linear Restrictions

Consider the “unrestricted” model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + e$$

Suppose we want to test the claim that only X_1 is relevant. Then we have $H_0 : \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$ vs $H_1 : H_0$ is not true. Consider the “restricted” model

$$Y = \beta_0 + \beta_1 X_1 + e$$

Generate the test statistic

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(N - K - 1)}$$

where SSR_r is the sum of squared residuals for the restricted model, SSR_{ur} is the sum of squared residuals for the unrestricted model, q is the number of restrictions, K is the number of regressors in the unrestricted model. As usual, we reject if $p - Value < \alpha$.

Caution About Gaussianity

Suppose your client wants to know if the the impact of advertising spending X upon revenues Y varies in their two locations. They provide you with data from their North Carolina store as well as their New York store.

Suppose you conduct your EDA and find that none of the data appears Gaussian. Can you still “run” a regression?

Yes. 1) Our estimates will not be impacted (*ceteris paribus*), but 2) Our inference might be disrupted. Gaussianity is not required for OLS. The large sample properties of our estimators will approach Gaussian behavior as long as a few technical assumptions are met (e.g. zero expected error, orthogonality of regressors and errors, iid data, and finite fourth moments).

Gaussian data is a convenience, especially in small samples.

Caution About Subsamples

Suppose your client wants to know if the the impact of advertising spending X upon revenues Y varies in their two locations. They provide you with a dataset from their North Carolina store. They also provide a separate dataset for their New York store.

A common approach is to run a standard regression on the NC data, and a separate regression on the NY data.

This provides differential marginal effects. However, we can't statistically compare the marginal effects.

Caution About Subsamples

Instead we could run the regression $Y = \alpha + \beta X + \gamma D_{NC} + \delta X * D_{NC} + e$.
To answer the client's question we can run the test $H_0 : \delta = 0$.

We can view advertising spending X as the treatment and compute the treatment effect as $\frac{\partial Y}{\partial X}$

We can then compute the differential marginal effect as

$$\frac{\partial Y}{\partial X} |_{D_{NC}=1} - \frac{\partial Y}{\partial X} |_{D_{NC}=0} = \beta + \delta - \beta = \delta$$

Q: What does it mean if we fail to reject?

A: There is no difference in the marginal impact of advertising (X) on revenues (Y) for the NY and NC stores.

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$$Y_i = \alpha + \beta X_i + \gamma Z_i + e_i$$

Many difficulties arise when designing, implementing, and interpreting a regression model. In this lecture we will focus on a few of the most common problems you will face in practice.

For each “problem” we will highlight

- ① Consequence
- ② Detection
- ③ Solution

Common Problems When Implementing Regressions

Problem	Consequence	Detection	Solution
Include Irrelevant Regressors			
Heteroscedasticity			
Multicollinearity			
Perfect Multicollinearity			
Non-linearity in Regressors			
Non-linearity in Parameters			
Omit Relevant Regressors			

Many consequences of these problems are possible, but we can group into two broad categories:

- 1 Bias: Sample estimator does not equal the true parameter value. $E[\hat{\beta}] \neq \beta^0$, where β^0 is the true object of interest, and $\hat{\beta}$ is the OLS estimator. Under standard assumption, OLS estimators are unbiased.
- 2 Precision: All estimators have variability, reflecting uncertainty in our model. Usually measured via the standard error of the estimator $se(\hat{\beta})$. If this value is incorrect, then inference is wrong, and decisions may be impacted.

Common detection methods

- Visual inspection during EDA
- Formal hypothesis testing
- Subject matter expertise before regression is implemented

Each problem has myriad potential solutions. Generally speaking, they fall into two broad categories:

- Redefine the model BEFORE implementation
- Fix the estimates/standard errors AFTER implementation

No solution is perfect.

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Too many X / Too few N

Model	Fit	Errors
$Y = \alpha + \beta_1 X_1 + e$	Good	Small
$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + e$	Better	Smaller
$Y = \alpha + \beta_1 X_1 + \dots + \beta_{200} X_{200} + e$	Best	Smallest

Q: What's wrong with this logic?

A:

- 1 Spurious Regression Fit
- 2 Insufficient Degrees of Freedom

Let's look at each of these issues in turn.

To investigate the issue of Spurious Regression Fit, we need to first establish methods of measuring model fit.

Overall Fit via R^2

The variation of our outcome variable $s_Y^2 = \frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})^2$. Notice that this is built from the total variation of Y : $\sum_{i=1}^N (Y_i - \bar{Y})^2$
Useful decomposition of the total variation in Y_i :

$$TSS = ESS + SSR$$

- TSS (Total Sum of Squares) measures the total sample variation in Y_i as $\sum_{i=1}^N (Y_i - \bar{Y})^2$
- ESS (Explained Sum of Squares) measures the sample variation in $\hat{Y}_i$ as $\sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2$
- SSR (Sum of Squared Residuals) measures the sample variation in residuals $\hat{e}_i$ as $\sum_{i=1}^N \hat{e}_i^2$

Overall Fit via R^2

From this decomposition we can build

$$R^2 = \frac{ESS}{TSS} = \frac{\sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} = 1 - \frac{\sum_{i=1}^N \hat{e}_i^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} = 1 - \frac{SSR}{TSS}$$

R^2 is the fraction of the total sample variation in Y that is explained by the model. Given this definition, R^2 lies between 0 and 1.

Example: $R^2 = .64$ means that 64% of the variation in Y can be explained by our model.

Overall Fit via R^2

Although we won't show, we can present R^2 as

$$R^2 = \frac{ESS}{TSS} = \frac{\sum_{i=1}^N (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} = \left[\frac{Cov(\hat{Y}_i, Y_i)}{\sigma_{\hat{Y}_i} \times \sigma_{Y_i}} \right]^2 = \left(\rho_{\hat{Y}_i, Y_i} \right)^2$$

R^2 can also be interpreted as the squared correlation coefficient between Y and $\hat{Y}$.

Spurious Regression Fit

Recall why we introduced R^2

Model	Fit	Errors
$Y = \alpha + \beta_1 X_1 + e$	Good	Small
$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + e$	Better	Smaller
$Y = \alpha + \beta_1 X_1 + \dots + \beta_{200} X_{200} + e$	Best	Smallest

Q: What's wrong with this logic?

A:

❶ **Spurious Regression Fit** (e.g. Overfitting)

❷ Insufficient Degrees of Freedom

$$R^2 = 1 - \frac{\sum_{i=1}^N \hat{e}_i^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} = 1 - \frac{SSR}{TSS}$$

- Adding regressors will never cause e to rise
- Adding irrelevant regressors may even shrink $\hat{e}$ (SSR)
- R^2 will never fall when adding more regressors, and it may even rise spuriously.

Adjusted R^2

We can “adjust” the R^2 formula to accommodate for this shortcoming.

$$R^2 = 1 - \frac{SSR}{TSS}$$
$$\bar{R}^2 = 1 - \frac{N-1}{N-K-1} \frac{SSR}{TSS}$$

where $N = \#$ of observations and $K = \#$ of regressors.

- The term $\frac{N-1}{N-K-1}$ acts as a penalty for adding regressors to the model
- Adding another regressor may lower SSR, and thereby increase R^2 , ..
- ...but it will also increase the penalty by increasing K
- The $\bar{R}^2$ rises if the SSR effect outweighs the penalty

Overall Fit via F test

We can connect the R^2 value to a hypothesis test for overall goodness of fit.

$$Y_i = \alpha + \beta_1 X_{1,i} + \cdots + \beta_K X_{K,i} + e_i$$

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_K = 0$$

H_1 : at least one of the β values is not equal to 0

Define K as the number of regressors (i.e. number of restrictions), and we get

$$F_{K, N-K-1} = \frac{R^2/K}{(1-R^2)/(N-K-1)}$$

Degrees of Freedom

Recall why we introduced R^2

Model	Fit	Errors
$Y = \alpha + \beta_1 X_1 + e$	Good	Small
$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + e$	Better	Smaller
$Y = \alpha + \beta_1 X_1 + \dots + \beta_{200} X_{200} + e$	Best	Smallest

What's wrong with this logic?

- 1 Spurious Regression Fit
- 2 **Insufficient Degrees of Freedom**

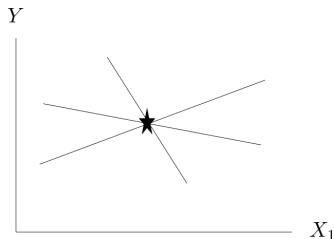
Notice that the term $N - K - 1$ is the Degrees of Freedom.

What if N is small and/or K is big?

Degrees of Freedom

Consider: $Y = \alpha + \beta_1 X_1 + e$

Q: Can we estimate the model with only one observation?

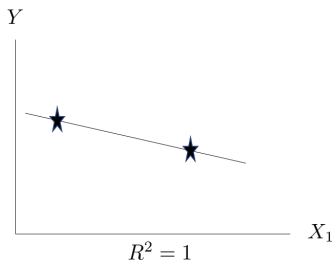
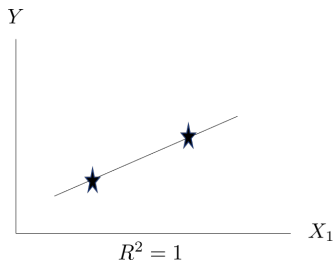


A: No. With only one observation an infinite number of regression lines are possible.

Degrees of Freedom

Consider: $Y = \alpha + \beta_1 X_1 + e$

Q: Can we estimate the model with two observations?

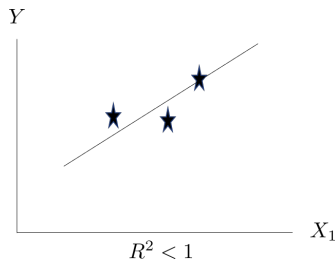


A: No. $df = N - K - 1 = 2 - 1 - 1 = 0$.
This isn't regression. This is a tautology.

Degrees of Freedom

Consider: $Y = \alpha + \beta_1 X_1 + e$

Q: Can we estimate the model with three observations?

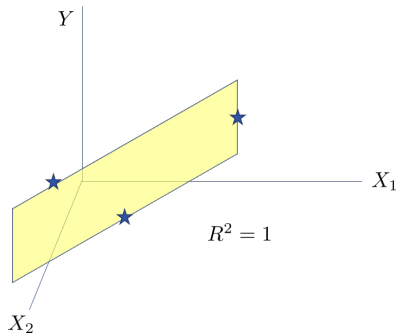


A: Yes. $df = N - K - 1 = 3 - 1 - 1 = 1$

Degrees of Freedom

Consider: $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + e$

Q: Can we estimate the model with three observations?

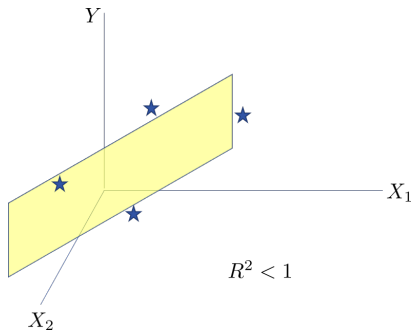


A: No. $df = N - K - 1 = 3 - 2 - 1 = 0$

Degrees of Freedom

Consider: $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + e$

Q: Can we estimate the model with four observations?



A: Yes. $df = N - K - 1 = 4 - 2 - 1 = 1$

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Defining Heteroscedasticity

Homoscedasticity

The error $\hat{e}$ has the same variance given any value of the regressors. In other words, the variance of the error is constant.

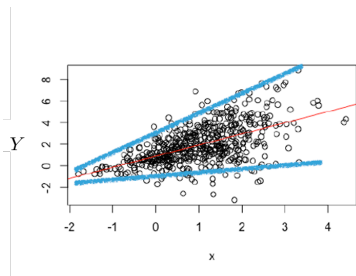
$$\text{Var}(e_i | X_{1,i}, X_{2,i}, \dots) = \sigma_e^2 \quad \forall i$$

Heteroscedasticity

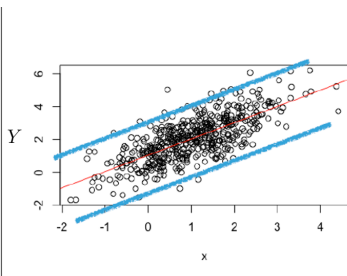
When $\text{Var}(e_i | X_{1,i}, X_{2,i}, \dots)$ depends on the regressors, $X_{1,i}, X_{2,i}, \dots$, the error term is said to exhibit heteroscedasticity (i.e. nonconstant variance)

$$\text{Var}(e_i | X_{1,i}, X_{2,i}, \dots) = \sigma_{ei}^2 \text{ where } \sigma_{ei}^2 \neq \sigma_{ej}^2$$

Detecting Heteroscedasticity

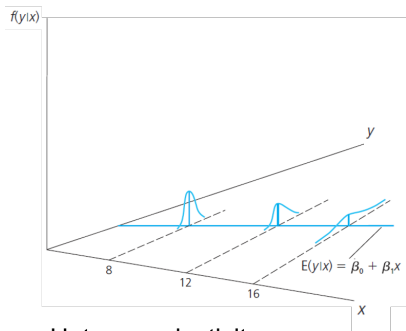


Heteroscedasticity
(Variance of errors
changes as X changes)

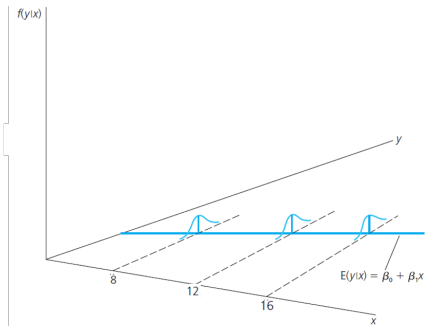


Homoscedasticity
(Variance of errors is
constant as X changes)

Detecting Heteroscedasticity

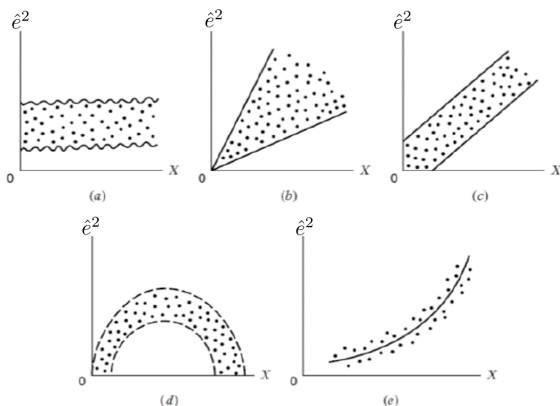


Heteroscedasticity
 $Var(e_i|X_i)$ increasing with X



The simple regression model
under homoscedasticity

Detecting Heteroscedasticity



Q: Which figures exhibit heteroscedasticity?

A: Homoscedastic: a || Heteroscedastic: b,c,d,e

Testing for Heteroscedasticity

Consider the following model, where all standard assumptions (other than homoscedasticity) apply

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \cdots + \beta_K X_{K,i} + e_i$$

- ➊ Estimate the model via OLS
- ➋ Obtain the residuals $\hat{e}_i$
- ➌ Estimate the auxiliary regression $\hat{e}_i^2 = \delta_0 + \delta_1 X_{1,i} + \cdots + \delta_K X_{K,i} + u_i$
- ➍ Obtain the $R_{\hat{e}^2}^2$
- ➎ Conduct the following test (Breusch Pagan “BP”)
 - $H_0 : \delta_1 = \cdots = \delta_K = 0$ (Homoscedastic)
 - $H_1 : \text{at least one of the } \delta \neq 0$ (Heteroscedastic)
 - $LM^{stat} = NR_{\hat{e}^2}^2$; $LM^{crit} = \chi_K^2$
 - If $LM^{stat} > LM^{crit}$ (or p-Value less than “size”) then reject $\rightarrow$ Heteroscedasticity is present

Consequence of Heteroscedasticity

- Coefficient estimates remain unbiased $E[\hat{\beta}] = \beta^0$.
- Standard Errors may be distorted. The usual OLS t -statistics don't follow t -distribution. Implies inference and decisions may be wrong.

Consequence of Heteroscedasticity

Distorted Standard Errors

$$\text{Recall } se(\hat{\beta}) = \frac{\hat{\sigma}_e}{\sqrt{SST_X(1-R_X^2)}}.$$

Heteroscedasticity might imply the $\hat{\sigma}_e$ are too high or too low.

Suppose, for instance, the estimated residual variance $\hat{\sigma}_e$ was overstated (i.e. the true residual variance is lower than our estimate)?

$\hat{\sigma}_e \uparrow \Rightarrow se(\hat{\beta}) \uparrow \Rightarrow t^{stat} \downarrow \Rightarrow \text{Type 2 Error} \uparrow$ (i.e. more likely to “Accept” a false null)

Q: If $H_0 : \beta_1 = 0$ vs $H_0 : \beta_1 \neq 0$, then what is the economic consequence of residual variance being overstated?

A: We are more likely to say that X_1 is unimportant, when it is in fact important to our model.

Solution for Heteroscedasticity

- This is NOT a problem that can be solved by increasing sample size.
- If we are lucky enough to know the form of the heteroscedasticity, there are ways to adjust the regression (e.g. GLS). However, that is rare in practice.
- Robust Standard Errors ala Huber White Eicker are the industry standard. In mixed-matrix form this is written as

$$\hat{\sigma}^2(\beta) = (X'X)^{-1} \sum_{i=1}^N e_i^2 x_i x_i' (X'X)^{-1}$$

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Defining Collinearity

Perfect Collinearity

Two regressors are perfectly collinear if there is an exact linear relationship between them: $X_1 = \lambda_1 + \lambda_2 X_2$

Perfect Multicollinearity

A set of regressors displays perfect multicollinearity if they share an exact linear relationship among them: $X_1 = \lambda_1 + \lambda_2 X_2 + \lambda_3 X_3$

(Multi)collinearity

A set of regressors display (multi)collinearity if they share a “strong” linear relation between them

Detecting Multicollinearity

- Check correlations among regressors. If “large” might have a problem.
- Check covariance matrix of estimated coefficients. If “large” might have a problem.
- Watch out for low t-stats combined with high F-stats for overall goodness of fit.
 - Recall $se(\hat{\beta}) = \frac{\hat{\sigma}_e}{SST_X(1-R_X^2)}$. If R_X^2 is high due to multicollinearity, the $se(\hat{\beta})$ might be high, t -stat low, Type 2 Error high
 - The F-test might be detecting that something in the model is important, but the similarity among the regressors makes it difficult for the t-tests to detect the source.
- Variance Inflation Factor (VIF)

Detecting Multicollinearity

Variance Inflation Factor

$$se(\hat{\beta}_k) = \frac{\hat{\sigma}_e}{\sqrt{SST_k(1 - R_k^2)}}$$

where $SST_k = \sum_{i=1}^N (X_{k,i} - \bar{X}_k)^2$ and R_k^2 is the R^2 from a regression of X_k on all other explanatory variables and a constant.

$$VIF = \frac{1}{1 - R_k^2}$$

Rule of Thumb:

If $VIF > 10$, then multicollinearity associated with X_k is “too high”

Perfect Multicollinearity

Suppose we estimate a factor model

$$R_{i,t} - R_{f,t} = \alpha + \beta R_{M,t} + \gamma R_{M,t}^{ppt} + e_t$$

where

- $R_{M,t}^{decimal}$ = market return in decimal form (e.g. 0.02)
- $R_{M,t}^{ppt}$ = market return in percentage points (e.g. 2)

Q: What's the problem here?

A:

$$R_{M,t}^{ppt} = 100 \times R_{M,t}$$

Therefore the regressors are perfectly collinear.

OLS cannot uniquely estimate β and γ .

Solution for Multicollinearity

- Do a better job choosing regressors during model design
- Gather more data
- Drop troublesome regressor
- Combine regressors using dimension reduction techniques (e.g. PCA)
- Use regularization methods that penalize large coefficients:
 - Ridge regression (L_2 penalty)
 - Lasso regression (L_1 penalty)
 - Elastic Net (combination of Ridge and Lasso)

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Non-linearities in Regressors

OLS is built for linearity in the parameters.

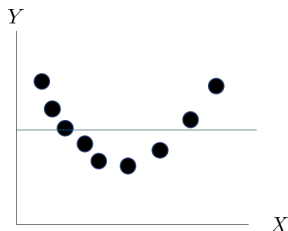
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + e$$

Models with non-linear regressors are still valid

$$Y = \beta_0 + \beta_1 X_1^2 + \beta_2 \exp(X_2) + e$$

Non-linearities in Regressors

Consider the following data



Q: Is a linear model $Y = \alpha + \beta X + e$ appropriate?

A: No. Perhaps altering the regressors is more appropriate?

$$Y = \alpha + \beta X^2 + e$$

Dollar vs Percentage Relationships

Suppose we model prices directly:

$$P_{i,t} = \alpha + \beta P_{M,t} + e_t$$

Q: Interpretation of β ?:

A: a \$1 increase in the market price increases the asset price by β dollars.

Q: What's the problem with that approach?

A: a \$1 move means different things depending on the price level.

- \$100 $\rightarrow$ \$101 = 1% change
- \$1000 $\rightarrow$ \$1001 = 0.1% change

Using Ln as Proxy for % Δ

Consider a simple model

$$Y = \alpha + \beta X + e$$

We know $\beta = \frac{\partial Y}{\partial X}$, or when discretized $\beta = \frac{\Delta Y}{\Delta X}$.

Consider a semi-log model

$$\ln(Y) = \alpha + \beta X + e$$

We can compute $\beta = \frac{\partial \ln(Y)}{\partial X}$. Using a chain rule we can show that this is equal to $\frac{1/Y \partial Y}{\partial X}$, which when discretized is $\frac{\Delta Y/Y}{\Delta X} \Rightarrow 100\beta \approx \frac{\% \Delta Y}{\Delta X}$. Note: This proxy is reasonable for small Δ values.

More precisely, note that $\widehat{\Delta \ln(Y)} = \hat{\beta} \Delta X$, from which we find

$$\% \Delta \hat{Y} = 100 \times [\exp(\hat{\beta} \Delta X) - 1]$$

Ln Transformations

The following are approximations. Exponentiate to get more exact results.

$Y = \alpha + \beta X$: For each 1 unit change in X, Y changes by β units

$\ln(Y) = \alpha + \beta X$: For each 1 unit change in X, Y changes by 100β %

$Y = \alpha + \beta \ln(X)$: For each 1% change in X, Y changes by $\beta/100$ units

$\ln(Y) = \alpha + \beta \ln(X)$: For each 1% change in X, Y changes by β %

Elasticity

The elasticity of Y with respect to X

$$\varepsilon_{YX} = \frac{\% \Delta Y}{\% \Delta X} = \frac{\Delta Y / Y}{\Delta X / X} \approx \frac{\partial Y}{\partial X} \frac{X}{Y}$$

Non-linearities in Parameters

OLS is built for linearity in the parameters.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + e$$

Models with non-linear parameters are invalid

$$Y = \beta_0 + \beta_1^2 X_1 + \ln(\beta_2) X_2 + e$$

Solution is to use nonlinear least squares or similar nonlinear methods, which are beyond our scope.

Outline

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- 2 Estimators
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- 3 Dummy Variables
- 4 Inference in Regression
- 5 Regression “Problems”
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 - Multicollinearity
 - Non-linearities
 - Problems Summary

Problem	Consequence	Detection	Solution
Include Irrelevant Regressors	Unbiased; se wrong; R^2 may be inflated	Theory; drop and check for change in overall goodness of fit	Drop regressor
Hetero.	Unbiased; OLS standard errors are incorrect	Plot Var(residual) vs combinations of X's; Goldfeld Quandt; BP; White	GLS, FLGS, Hetero. robust standard errors
Multicollinearity	Unbiased; standard errors incorrect	X correlations; $VIF > 10$; Low t-stat with high F-stat; drop regressors and detect changes in estimates	Drop a regressor; combine regressors; joint hypothesis test
Perfect Multicollinearity	OLS invalid	Model design; X correlations	Drop a regressor
Non-linearity in Regressors	Parameter interpretation might be wrong	Model Design	Redefine regressors
Non-linearity in Parameters	OLS is invalid	Model design	Nonlinear Least Squares (beyond our scope)
Omit Relevant Regressors	Bias; Altered Interpretation	Theory; Marginal effect of interest changes when augment regressors	Proxies; FE; IV

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